OSINACHI CHUKWUJAMA

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PROFILE

Photonics master's student with a First Class degree in Mechanical Engineering, working at the meeting point of optics, imaging and machine learning. I have derived and implemented diffusion and flow-matching models in PyTorch, deployed vision models on embedded GPUs, worked hands-on with spatial light modulators, and built optics simulations. I am looking for a summer research project in computer vision, computational imaging or generative modelling.

RESEARCH INTERESTS

Computational imaging, generative models for vision (diffusion, flow matching), image restoration and reconstruction, optimal transport in machine learning.

EDUCATION

Erasmus Mundus Joint Master, Intelligent Photonics (iPSRS) September 2026 – 2028
Université Jean Monnet (France), University of Eastern Finland, Université Paris-Est Créteil
Relevant coursework: Fourier optics, digital image processing, interferometry, probability and signal analysis, scientific computing with Python.

Bachelor of Engineering, Mechanical Engineering November 2017 – August 2023
Federal University of Technology, Owerri (FUTO), Nigeria. CGPA: 4.56/5.00 (First Class),
96th percentile in the department; best graduating student in Industrial and Production Engineering.

RESEARCH AND LABORATORY EXPERIENCE

Generative Modelling from First Principles 2026
Self-directed research

- Derived and implemented DDPM (forward/reverse process, ELBO, reparameterisation) and OT-conditional flow matching in PyTorch, covering training and sampling end to end.
- Studied the links between score matching (denoising and implicit variants, NCSN), flow matching and normalising flows; wrote a forward-diffusion noising tool in Go for synthetic data generation.
- Proposed a comparison of OT-CFM and DDPM on photonics/laser data (TPU Research Cloud application).

Spatial Light Modulator (SLM) Laboratory Practical 2026
iPSRS M1 coursework *Université Jean Monnet*

- Generated diffraction gratings and Fresnel lens phase patterns in MATLAB and displayed them on an SLM.
- Observed and analysed the resulting diffraction patterns, and computed the gratings' spatial frequencies from measurements.

Surveillance Drone with Real-Time Object Detection August 2023 – March 2024
Undergraduate final project and follow-up, supervised by Prof. O.O. Obiukwu *MEE, FUTO*

- Collected and annotated a three-class dataset, fine-tuned YOLOv5 (70% mAP) and ran live inference on a drone's camera stream via ONNX.
- Deployed the model on an NVIDIA Jetson Nano with TensorRT, cutting processing time by 44%.

Optics Playground In progress
Personal project, pre-zeemax.vercel.app

- Interactive 2D ray-tracing simulator with finite-size lenses, being extended into a Fourier optics bench.

SKILLS

Machine learning	PyTorch: DDPM, flow matching, score matching, CNN/ResNet training, transfer learning; ONNX, TensorRT
Mathematics	Probability, variational inference (ELBO), optimal transport for flow matching, Fourier analysis
Programming	Python (proficient), MATLAB, Go, TypeScript/JavaScript; Git, Linux, CI/CD, automated testing
Languages	English: C1 (IELTS 8.0)

WORK EXPERIENCE

Dstrct Group March 2025 – May 2026
Backend Engineer / Technical Lead *Lagos, Nigeria*

· Technical lead on an enterprise platform with a typed backend of nearly 1,000 files; worked directly with clients to turn requirements into engineering tasks.

Hotels.ng June 2024 – January 2026
Software/DevOps Engineer (part-time) *Lagos, Nigeria*

· Built a concurrent Python/Celery application distributing AI workloads over RabbitMQ; wrote CI/CD and monitoring tooling.

PUBLICATIONS

Quantum Computing in Artificial Intelligence: A Review of Quantum Machine Learning Algorithms, Path of Science. doi:10.22178/pos.117-25

Secure and Resilient Industrial IoT Architectures for Smart Manufacturing: A Comprehensive Review, JERR (2025). doi:10.9734/jerr/2025/v27i61548

Clinical Applications of 3D-Printed Medical Implants, JOPIR. doi:10.69613/22dn3m33

HONOURS, TEACHING AND COMMUNICATION

PTDF Scholarship (2018) for academic excellence.

Microsoft Certified Trainer and former **Microsoft Learn Student Ambassador**: trained over 300 students in software engineering and cloud technologies.

Director of Research, Mechanical Engineering, FUTO: ran hands-on training in machine learning, microcontrollers and 3D printing for 50+ students and staff. Co-founder of Hack4FUTO.

Technical writer at vicradon.hashnode.dev.